25901580 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCZGY202 - FUNCTIONAL ZOOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. ---------- catalyses the conversion of fibrinogen to fibrin. [K] / [CO1]

   a). Thrombin b). Prothrombin

   c). Calcium ions d). Vitamin K

2. HIV, the virus that causes AIDS, primarily infects and destroys [U] / [CO4]

   a). B-lymphocytes b). CD8-positive T-cells

   c). CD4-positive T-cells d). Erythrocytes

3. Which class of immunoglobulin is predominant in mucosal secretions? [K] / [CO4]

   a). IgG b). IgA

   c). IgM d). IgE

4. Addison’s disease is due to hyposecretion of _____. [K] / [CO3]

a). Cortisol and aldosterone b). Adrenaline

c). Growth Hormone d). Thyroid Stimulating Hormone

5. Renin is secreted by [U] / [CO1]

a). Liver b). Juxta glomerular cells

c). Adrenal cortex d). Posterior pituitary

6. Name the disease caused by the intake of low quality and quantity of dietary proteins. [U] / [CO1]

a). Obesity b). Kwashiorkor

c). Rickets d). Scurvy

7. Hormones act by binding with _____. [K] / [CO3]

a). Enzymes b). DNA

c). Ribosomes d). Receptors

8. The immune cells responsible for antigen presentation include [U] / [CO4]

a). Dentritic cells b). RBCs

c). Platelets d). Muscle cells

9. A major advantage of recombinant vaccines is [U] / [CO4]

a). Risk of causing disease b). High contamination risk

c). Better safety profile d). Requires live pathogen

10. The main feature that distinguishes acquired immunity from innate immunity is: [K] / [CO4]

a). Presence of physical barriers b). Non-specific response

c). Immunological memory d). Immediate action

11. The action potential spreads to the interior of muscle fibre through: [K] / [CO2]

a). Sarcolemma b). T-tubule

c). Sarcoplasm d). None of the above

12. Fight or flight response involves _____. [U] / [CO3]

a). Estrogen b). Adrenal medulla hormones

c). Calcitonin d). GH

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. Explain briefly hybridoma technology. [U] / [CO4]

14. Differentiate between conduction in myelinated and non myelinated nerve fibers. [U] / [CO2]

15. State the role of bone marrow in immunity. [K] / [CO4]

16. What is meant by Hamburger phenomenon? [K] / [CO1]

17. Differentiate between axon and dendrite. [K] / [CO2]

18. Explain the role of TRH and its regulation. [U] / [CO3]

19. Explain how the structure of the renal corpuscle is suited for filtration. [U] / [CO1]

20. What is troponin? [K] / [CO2]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Compare autoantigens and alloantigens with suitable examples. [K] / [CO4]

22. Describe different types of WBCs. [U] / [CO1]

23. Describe mechanism of antigen- antibody reaction. [U] / [CO4]

24. Outline the causes and consequences of hypoxia and hypercapnia. [U] / [CO1]

25. Give an account on rigor mortis. [U] / [CO2]

26. Write an account on neural disorders in man. [K] / [CO2]

[5x4 = 20]